

Exam Date & Time: 15-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Pattern classification [NPT_4147]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A biometric security system must identify individuals using fingerprints.
Analyze how different pattern recognition approaches (statistical vs. structural) influence the **accuracy and computational efficiency** of fingerprint authentication. (5)
A) **Explain** which approach would be more suitable for large-scale systems and why.
B) Given a dataset of handwritten digits, Illustrate feature extraction approach that captures both structural and statistical features. (5)
- 2) For a weather forecasting system, **examine** how **Bayesian inference** combines **prior probabilities** and **likelihood functions** to classify complex weather patterns. (5)
Analyze how updates in prior probabilities (e.g., seasonal trends) can influence classification outcomes and prediction reliability.
A) (5)
B) A medical image classifier uses binary features (tumor present/absent). Illustrate how Bayes Decision Theory applies to binary classification. (5)
- 3) A machine learning model predicts air pollution levels in a city.
Evaluate how **maximum likelihood estimation (MLE)** helps in identifying optimal model parameters under noisy environmental data. (5)
A) **Compare** MLE with other estimation methods (e.g., least squares) in terms of **robustness and interpretability**.
B) You are designing a face recognition system that must differentiate between 10 people. Illustrate how Linear Discriminant Analysis (LDA) can be applied (5)
- 4) A handwriting recognition system employs a simple neural network model.
Analyze how the **perceptron criterion** influences the formation of the **decision boundary** between handwritten characters. (5)
A) **Discuss** the limitations of this approach in handling **non-linear separability** and how it can be improved.
B) A credit card fraud detection model applies a Hyperbox classifier. Illustrate how hyperboxes define class boundaries in multi-dimensional data. (5)
- 5) In a **speech emotion recognition** system, fuzzy logic is used to classify emotions such as happy, sad, or angry. (5)

- A) **Analyze** how **fuzzy rule-based systems** manage overlapping acoustic features like pitch and tone.
Evaluate their effectiveness compared to crisp classifiers in dealing with **ambiguous** or **mixed emotional expressions**.
- B) A gesture recognition system tracks hand movements over time. Illustrate how temporal pattern recognition is used to identify gestures. (5)

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Event Name	Subject
058-END SEMSTER-ODD-2025	NPT_4147:Pattern classification

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	NPT_4147:Pattern classification

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	25(50%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)